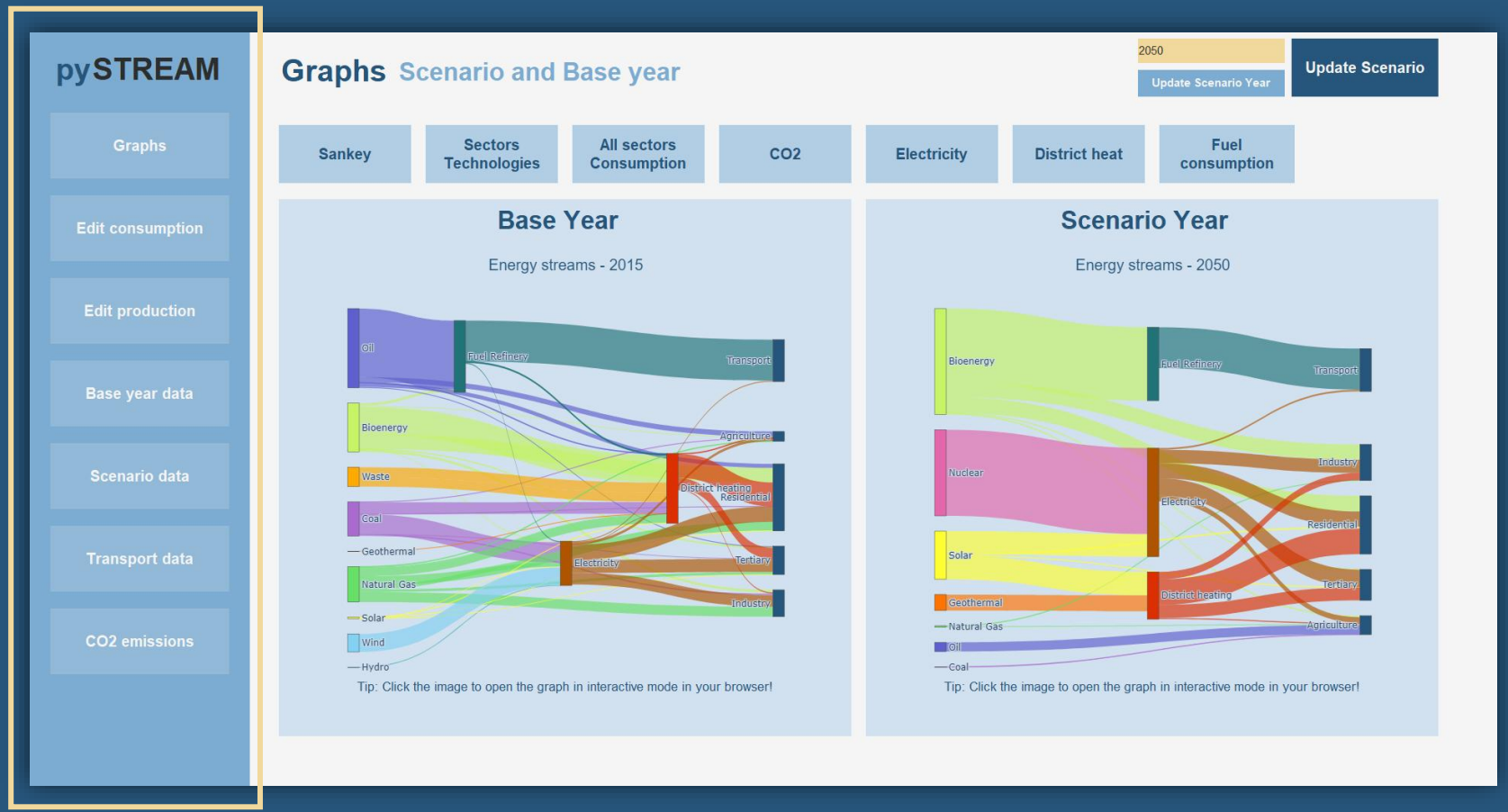
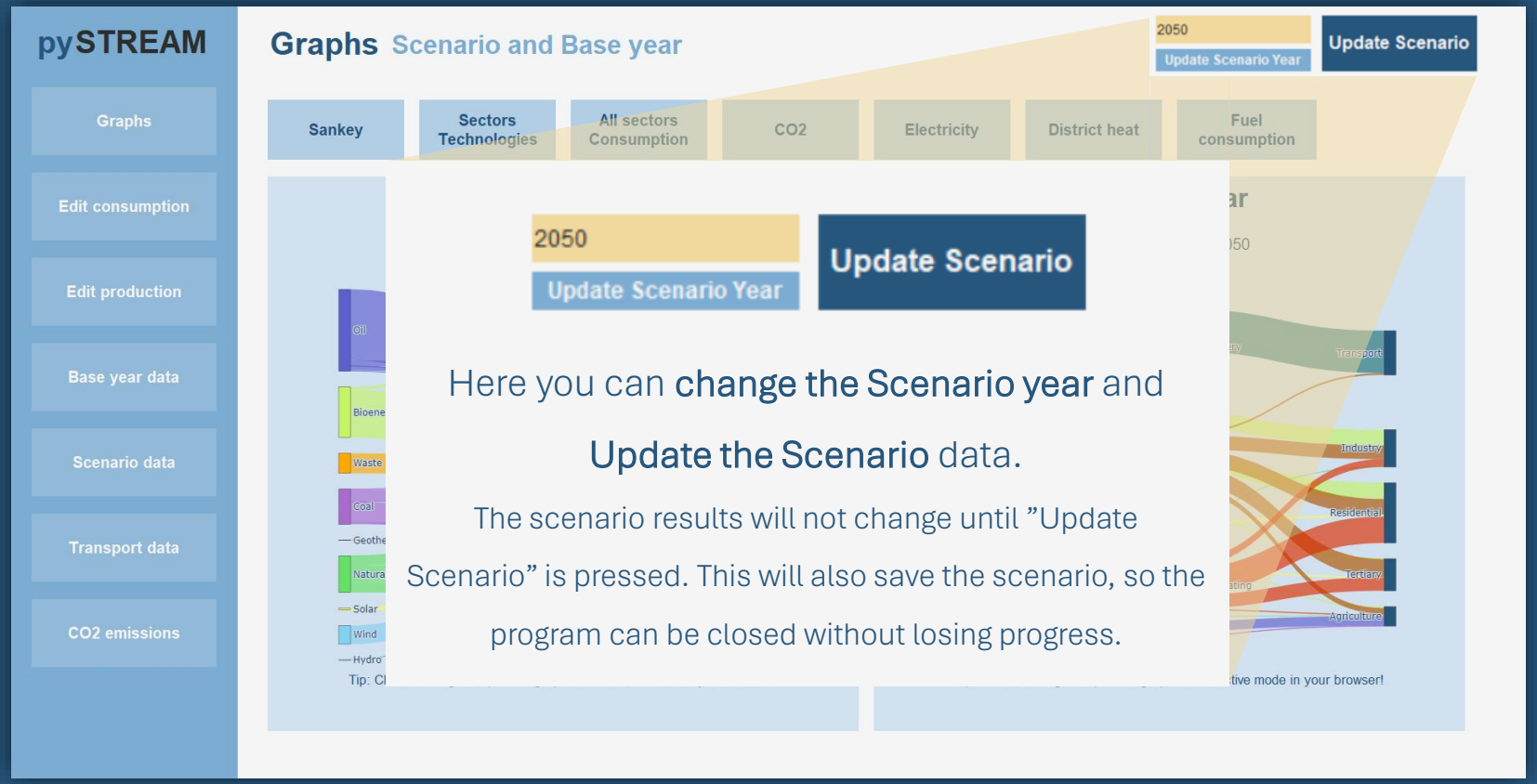
Welcome to pySTREAM!

This is your **start page**.

This is the **navigation menu.** Here you can switch to the different pages in pySTREAM.



Here you can **Change the Scenario year** and **Update the Scenario** data.



Yellow buttons like these appear next to tables where you can edit the data.

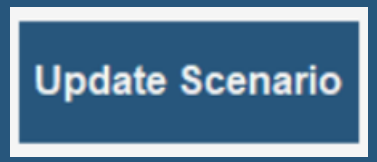


To edit the tables you can *Double-click* the cell you want to edit.

You need to press *Enter* to complete the editing of the cell. If you click away from the cell without pressing enter, the cell will go back to the original value.

When you are done editing the cells in the table, press the **yellow save button.**

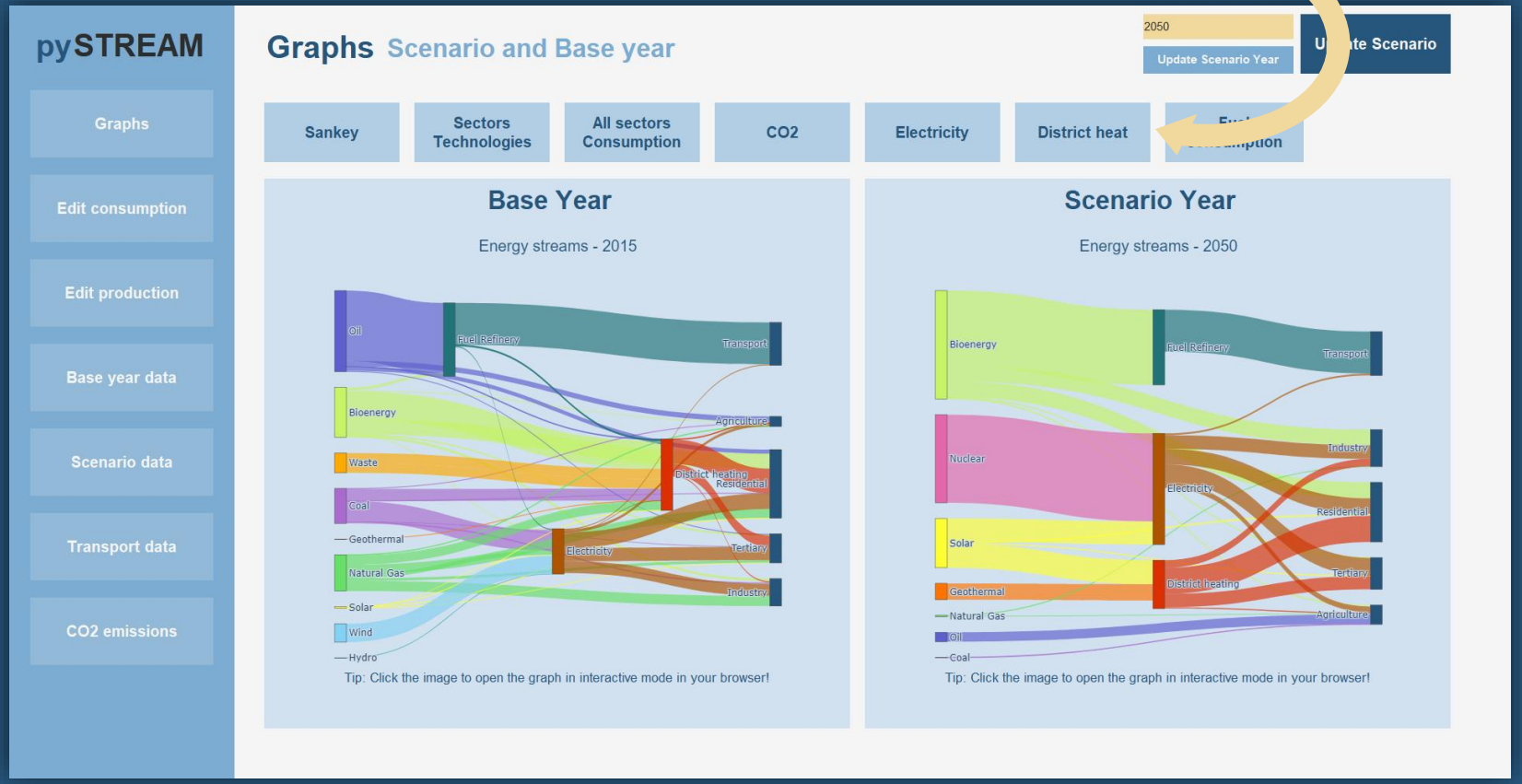
The Scenario will not completely update and save until you click the **blue Update Scenario button** in the top right corner.



When you press this button your changes will also be saved in the data sheet, so you can open and close the program without losing your work.

This is the **Graphs page** for **Scenario and Base year**.

You can switch between graphs with these buttons:



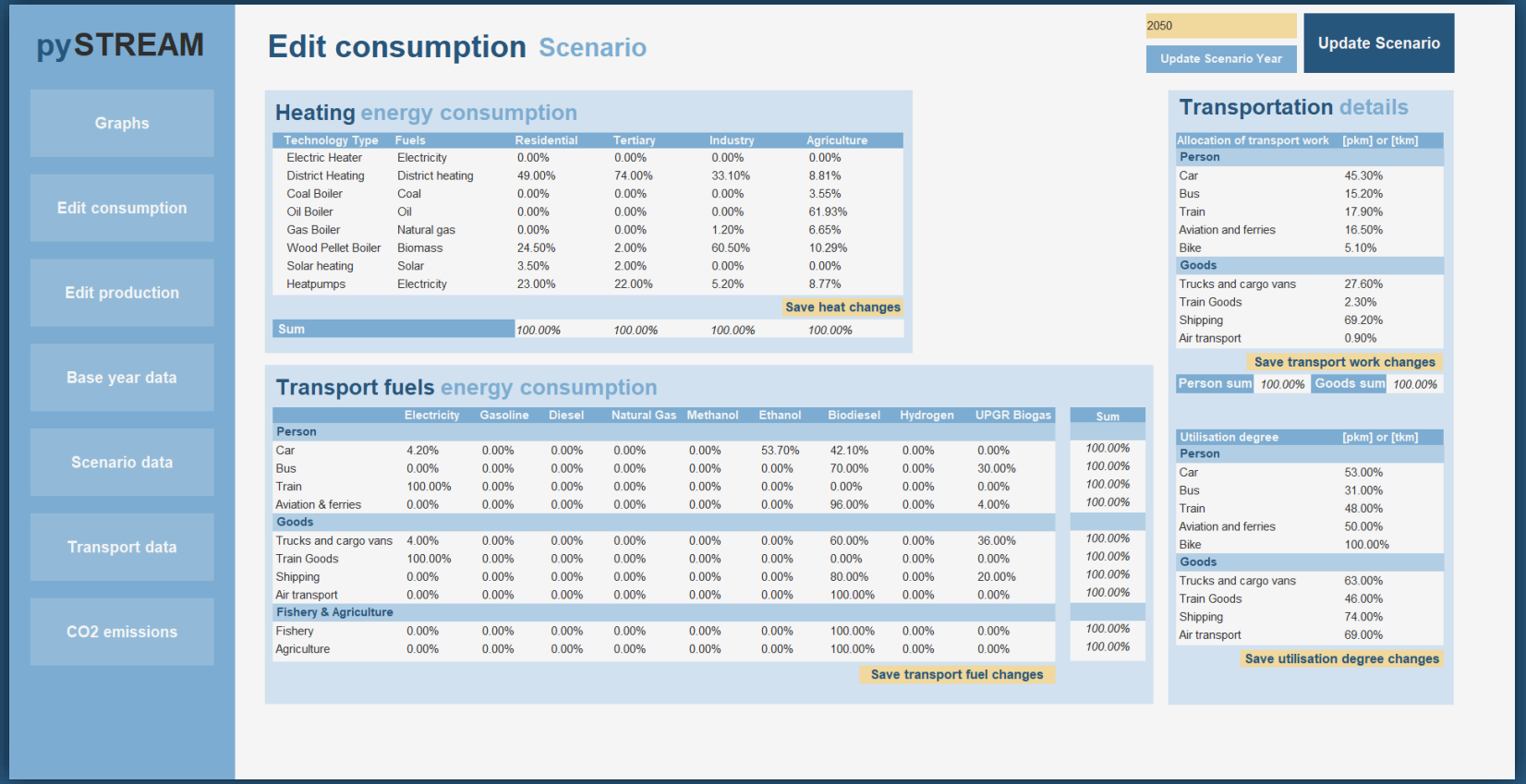
Here you can view different graphs of and analyze the results of your scenario by comparing with the base year.

This is the **Edit consumption** page for **the Scenario**.

Here you can edit the energy consumption

In **Heating energy consumption** you can edit where the heating in the different sectors should come from.

The sectors *Residential, Tertiary, Industry, and Agriculture* should each have a sum of 100%.



In **Transportation details** you can edit the **Allocation of transport work** and the **Utilisation degree.**

Allocation of transport is how often each transport type is used. The sum should be 100% for *Person* and *Goods* respectively.

Utilisation degree is how ”full” the transport types normally are, ie. a four-person car with two people in it has a utilisation degree of 50%.

In **Transport fuels energy consumption** you can edit which fuel types should be used for each transport type.

Person, Goods, and Fishery & Agriculture are the categories each transport type belong to.

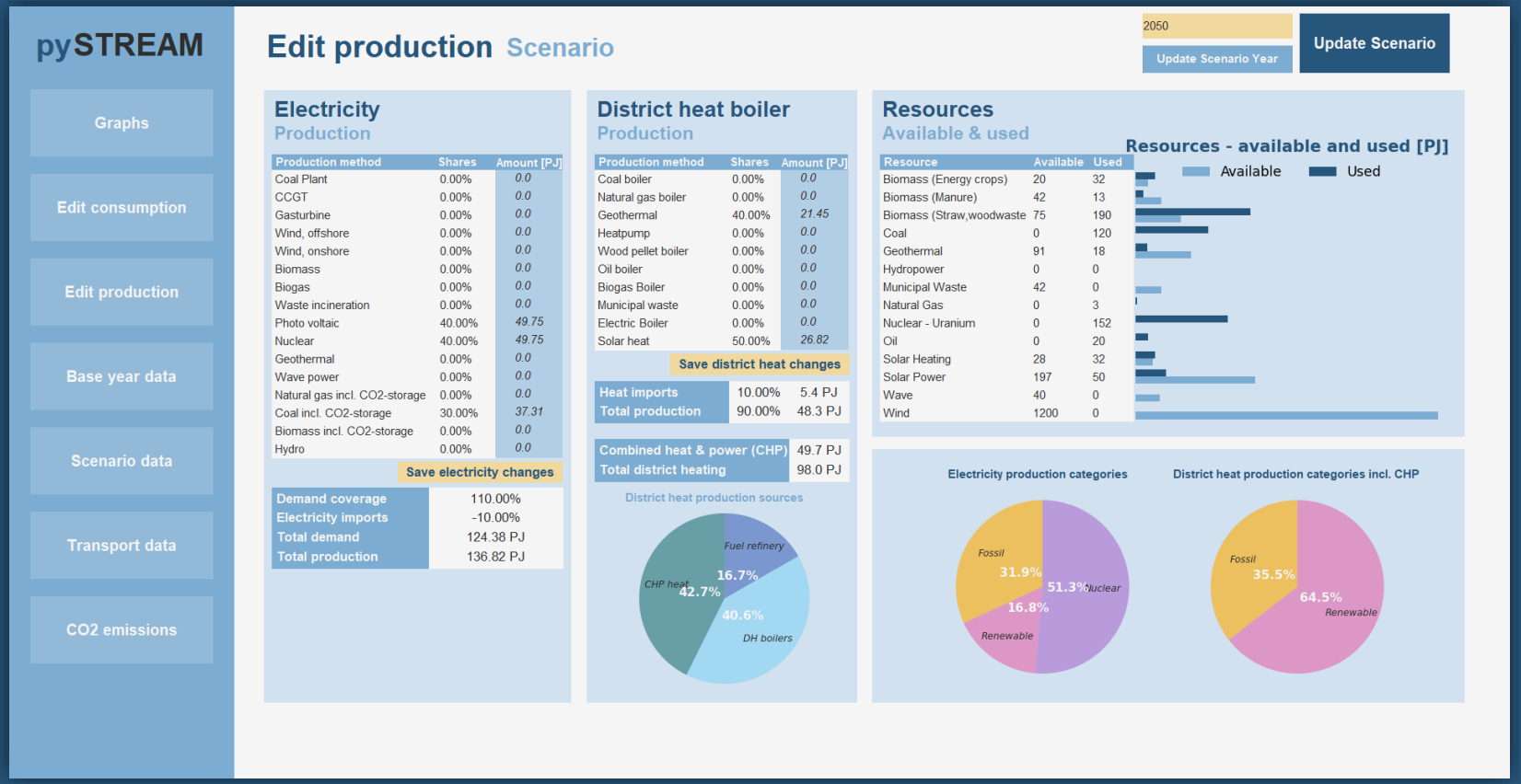
The transport types *Car, Bus, Train, Aviation & ferries, Trucks and cargo vans, Train Goods, Shipping, Air transport, Fishery, and Agriculture* should each have a sum of 100%.

This is the **Edit production** page for **the Scenario**.

Here you can edit how the required energy should be produced

In **Electricity production** you can edit which production methods should be used to meet the *electricity demands*.

Electricity imports will be positive (importing) if the sum of production is less than the demand (100%) and negative (exporting) if the sum of production is more than the demand (100%).



Ressources available & used shows you the amount of ressources the country possesses and how much the scenario is currently using.

If your scenario uses more of a resource than is available, the ressource will have to be imported.

In **District heat boiler production** you can edit which heating production methods should be used to meet the *district heating demands*.

Heat imports will be positive (importing) if the sum of production is less than the demand (100%) and negative (exporting) if the sum of production is more than the demand (100%).

CHP heat production is affected by the **Electricity production**, since only some power production methods also have heat production (Gas, Biomass, Waste, Coal).

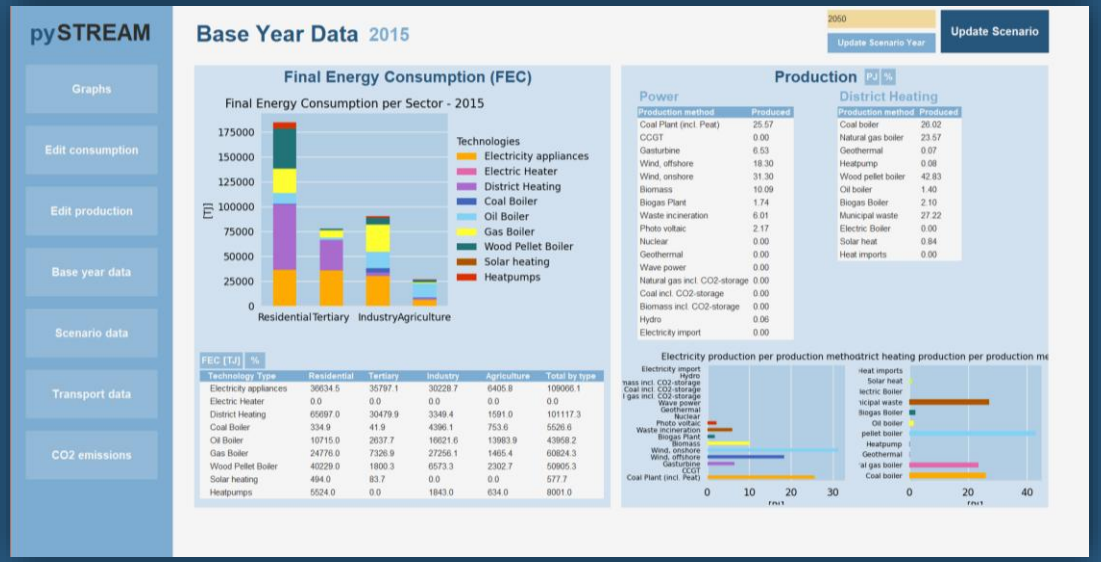
These are the **Base year data** and the **Scenario data**.

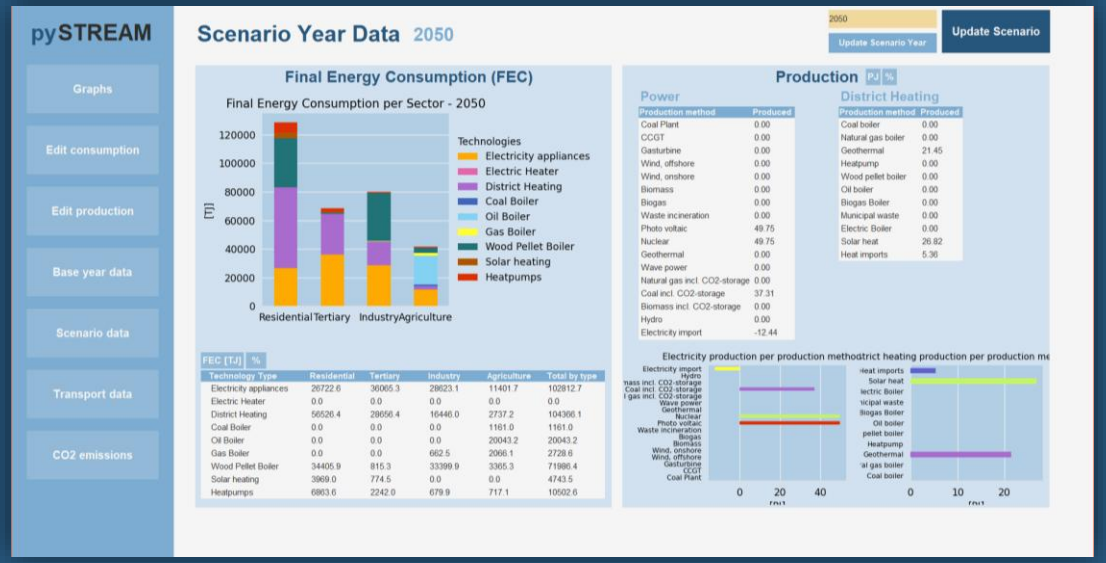
On these pages you can only view data from the two years.

The **Final Energy consumption** shows you the *electricity* and *heat* consumption of each sector.

In the table you can switch between viewing the data in TJ or in percentages.

Electricity appliances will always be 100% for all sectors, since they all recieve electricity from the grid.





Production of **Power** and **District Heating** shows you how the energy is produced.

In the table you can switch between viewing the data in PJ or in percentages

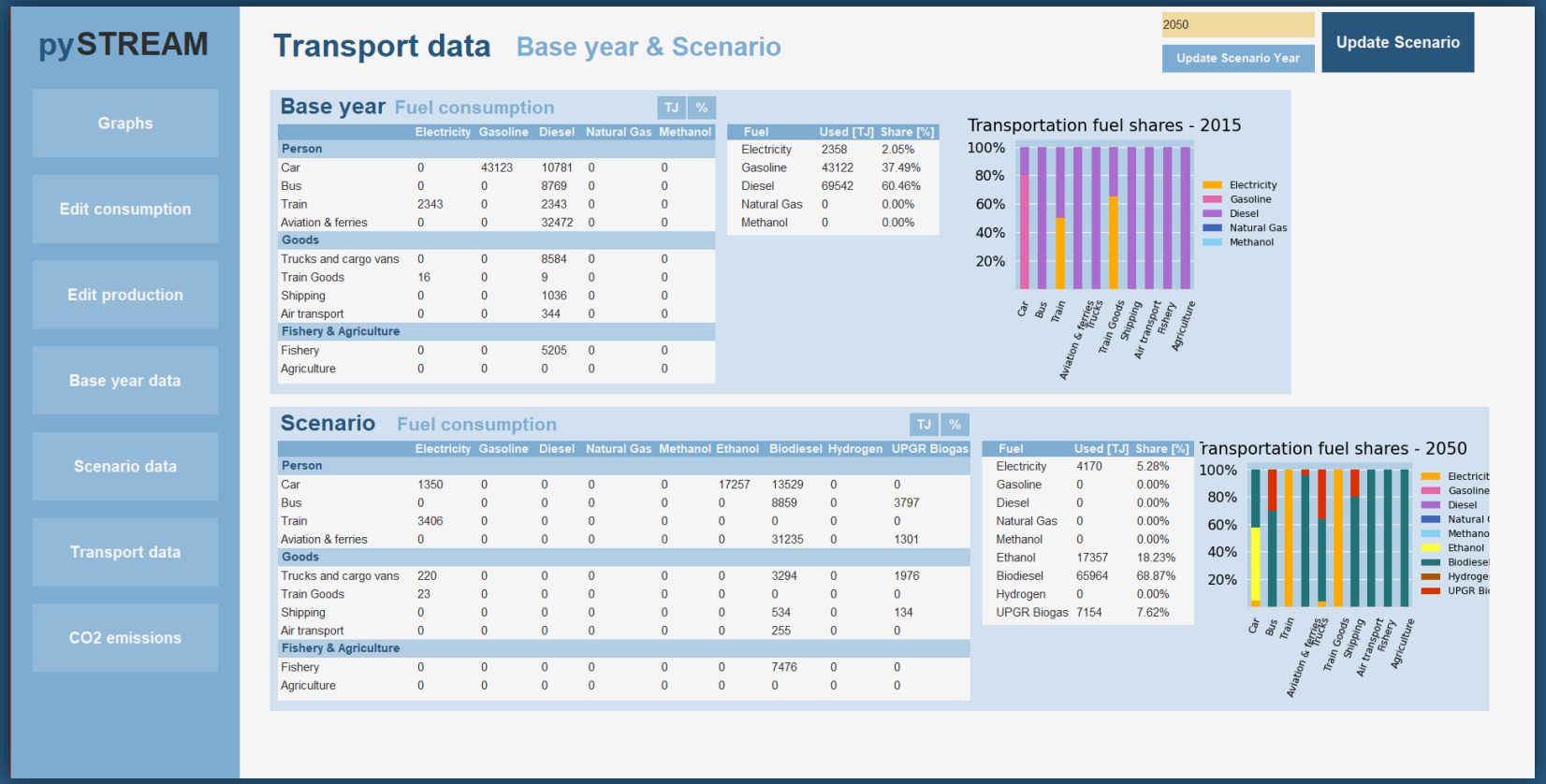
This is the **Transport data** for **Base year data** and the **Scenario data**.

On this pages you can only view data from the two years.

Both years have the same two tables

The first table contains a detailed view of the fuel consumption for all transport types and each of their fuel type consumption.

In the table you can switch between viewing the data in TJ or in percentages.



The second table contains the total amount consumed of each *transport fuel type* and the percentage of the total fuel consumption.

The base year has fewer transport fuel types.

This is the **CO2 data** for **Base year data** and the **Scenario data**.

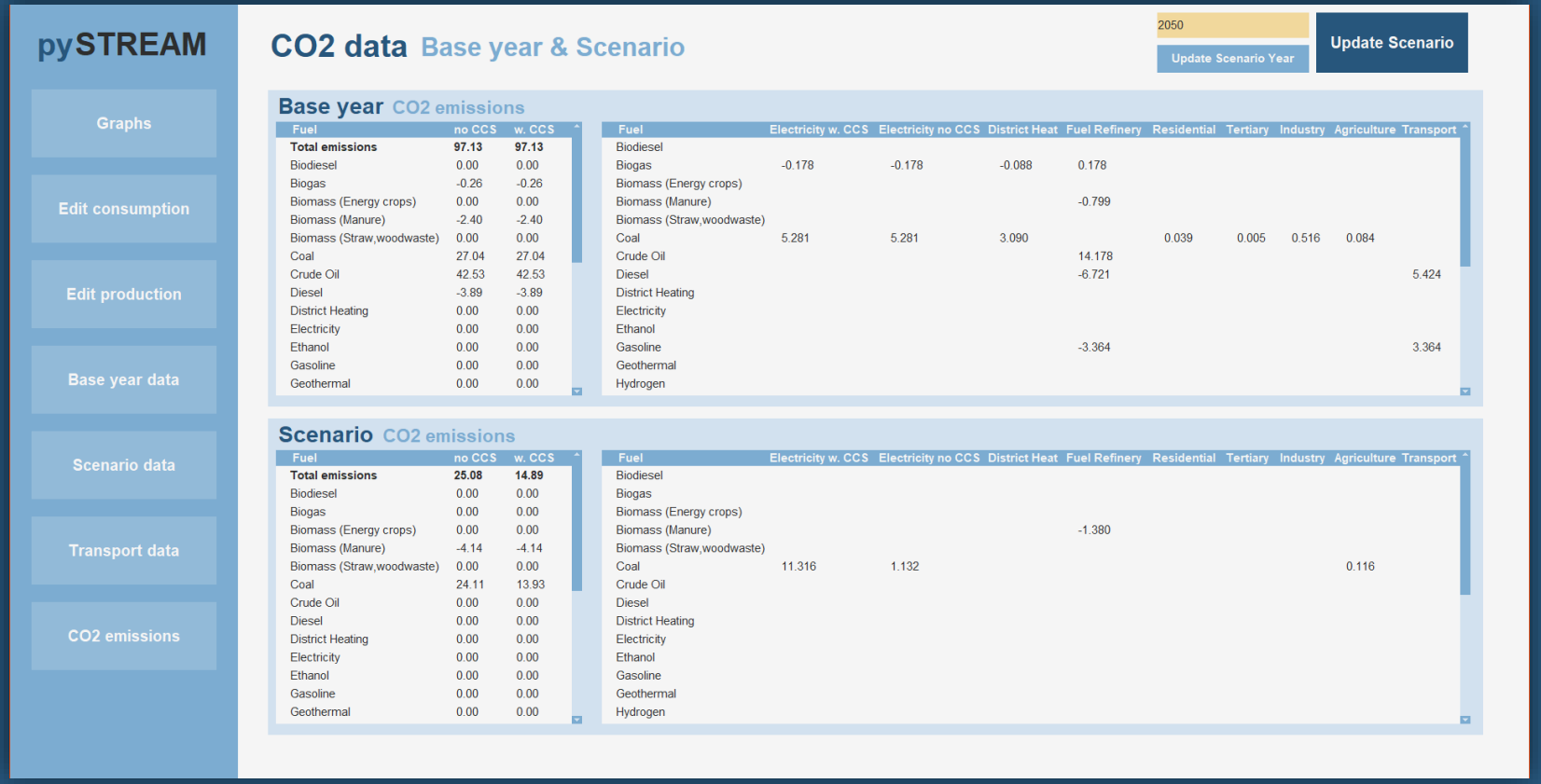
On this pages you can only view data from the two years.

Both years have the same two tables

The first table contains the total emissions from each *fuel type,* and the *total emissions,* with or without CCS.

CCS (Carbon Capture & Storage) is only available for Natural Gas, Coal, and Biomass.

In the **Edit production** page you can define if you want to use these technologies with CCS.



Base year CO2 emissions

Fuel	no CCS	w. CCS
Total emissions	**97.13**	**97.13**
Biodiesel	0.00	0.00
Biogas	-0.26	-0.26
Biomass (Energy crops)	0.00	0.00
Biomass (Manure)	-2.40	-2.40
Biomass (Straw,woodwaste)	0.00	0.00
Coal	27.04	27.04
Crude Oil	42.53	42.53
Diesel	-3.89	-3.89
District Heating	0.00	0.00
Electricity	0.00	0.00
Ethanol	0.00	0.00
Gasoline	0.00	0.00
Geothermal	0.00	0.00

Fuel	Electricity w. CCS	Electricity no CCS	District Heat	Fuel Refinery	Residential	Tertiary	Industry	Agriculture	Transport
Biodiesel									
Biogas	-0.178	-0.178	-0.088	0.178					
Biomass (Energy crops)									
Biomass (Manure)				-0.799					
Biomass (Straw,woodwaste)									
Coal	5.281	5.281	3.090		0.039	0.005	0.516	0.084	
Crude Oil				14.178					
Diesel				-6.721					5.424
District Heating									
Electricity									
Ethanol									
Gasoline				-3.364					3.364
Geothermal									
Hydrogen									

Scenario CO2 emissions

Fuel	no CCS	w. CCS
Total emissions	**25.08**	**14.89**
Biodiesel	0.00	0.00
Biogas	0.00	0.00
Biomass (Energy crops)	0.00	0.00
Biomass (Manure)	-4.14	-4.14
Biomass (Straw,woodwaste)	0.00	0.00
Coal	24.11	13.93
Crude Oil	0.00	0.00
Diesel	0.00	0.00
District Heating	0.00	0.00
Electricity	0.00	0.00
Ethanol	0.00	0.00
Gasoline	0.00	0.00
Geothermal	0.00	0.00

Fuel	Electricity w. CCS	Electricity no CCS	District Heat	Fuel Refinery	Residential	Tertiary	Industry	Agriculture	Transport
Biodiesel									
Biogas									
Biomass (Energy crops)									
Biomass (Manure)				-1.380					
Biomass (Straw,woodwaste)									
Coal	11.316	1.132						0.116	
Crude Oil									
Diesel									
District Heating									
Electricity									
Ethanol									
Gasoline									
Geothermal									
Hydrogen									

The second table contains a detailed view of consumption of each fuel type and in which sector the emission occurs.

The sectors here include *Electricity production, District heat production, and Fuel refinery,* that are not final destinations of the energy. The sectors *Residential, Tertiary, Industry, Agriculture, and Transport* are the final energy destinations

The *Fuel refinery* converts primary fuels like *Biomass* or *Oil* into secondary fuels like Ethanol or Diesel, that can be used in sectors like *Transport*.

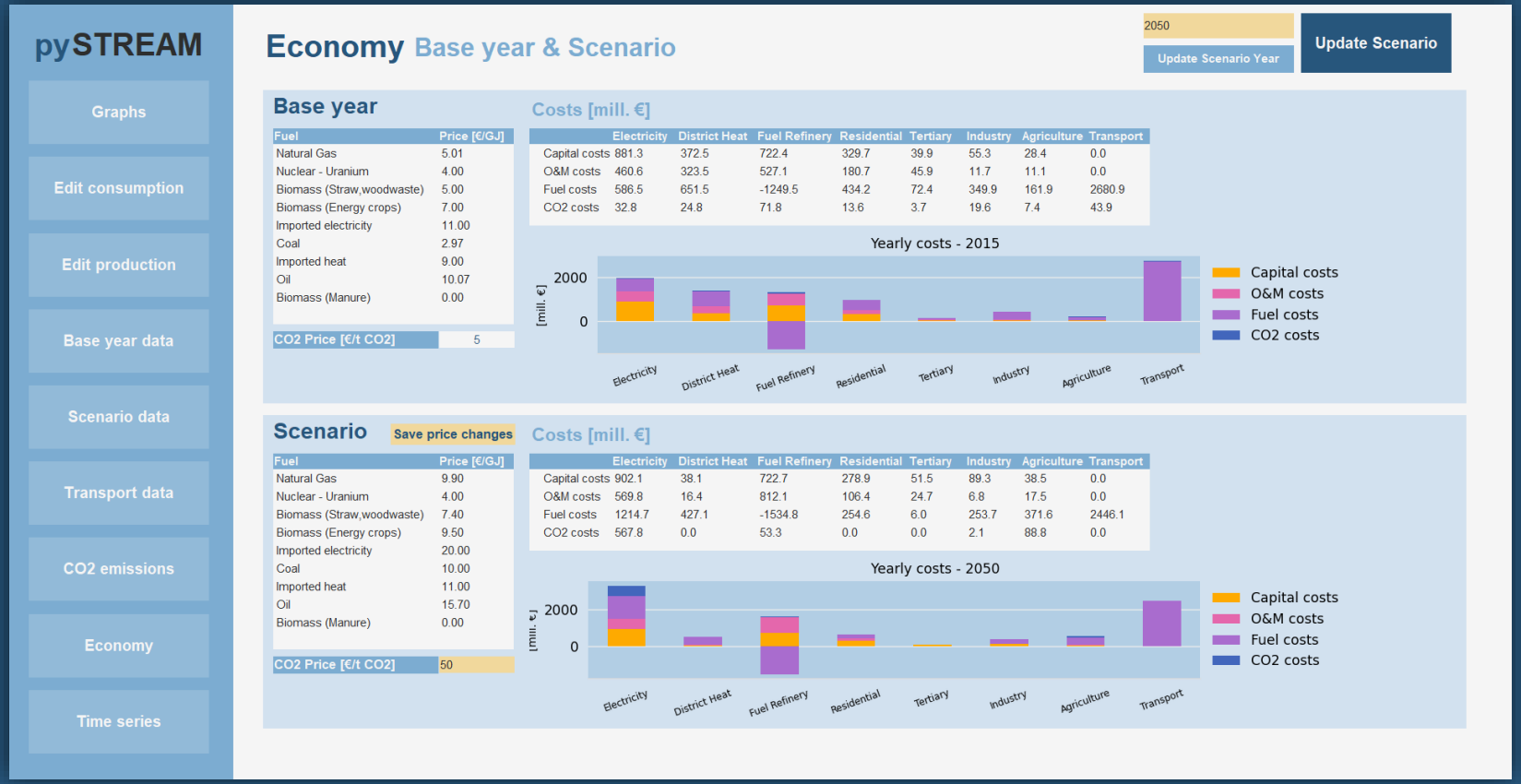
This is the **Economy data** for **Base year data** and the **Scenario data**.

On this pages you can edit the fuel prices and CO2 price for the **scenario year**,

and view different costs for each year for each sector.

The **fuel prices table** includes primary fuels that have editable prices.

You can also edit the **CO2 price**, which is a fixed price that is multiplied with the emissions for each technology.

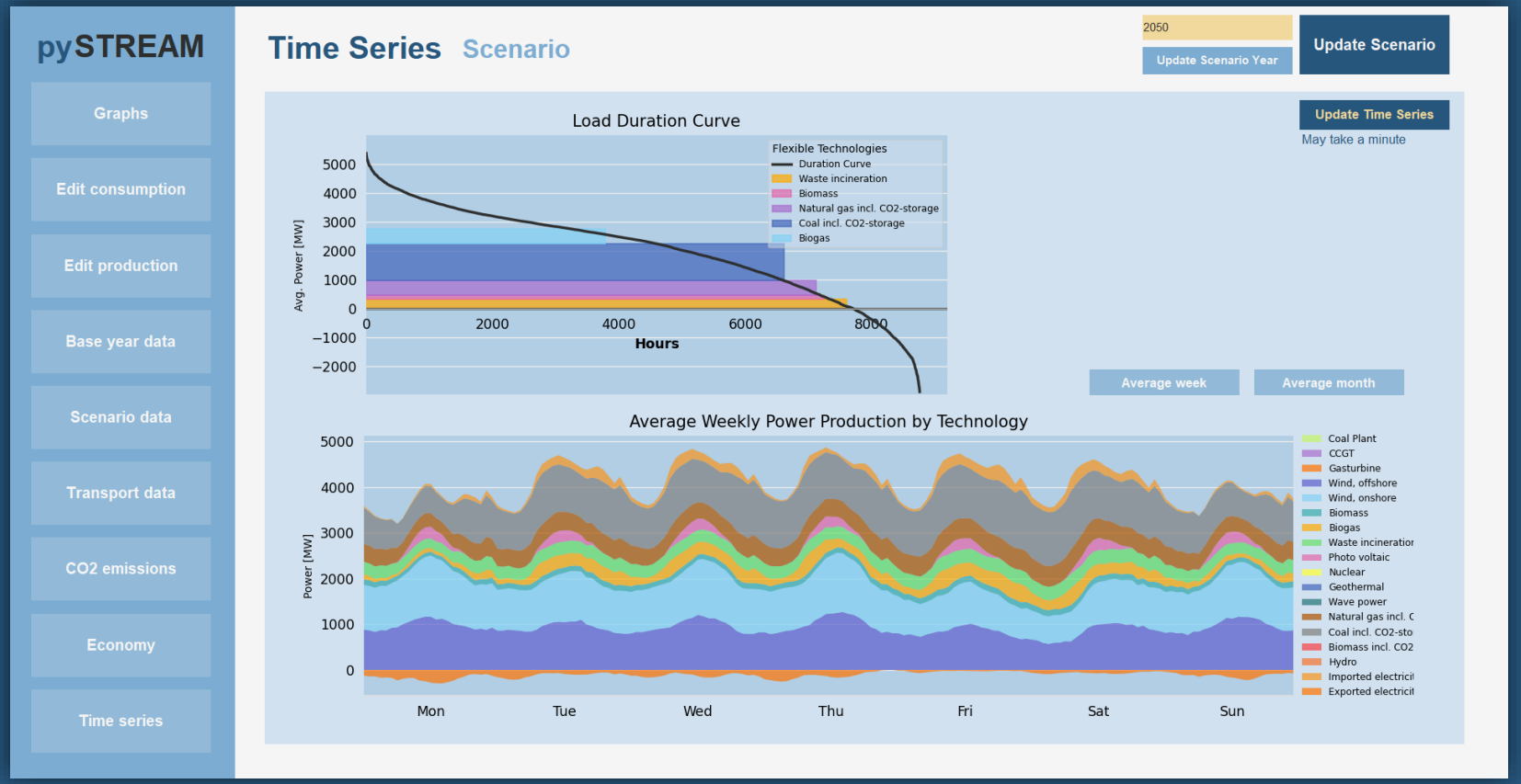


This is the **Time series** for the **Scenario**.

On this pages you can inspect the hourly production as a **Load Duration Curve** and **Average Weekly Power Production**.

The **Load Duration Curve** plot shows the electricity consumption per hour after intermittent renewable power has been subtracted, sorted from most to least.

Each dispatchable/flexible technology is plotted as the amount of hours it is actively producing (width) and the average power production (height).



The time series is updated by clicking the **"Update Time Series"** button.

The time series will not update with the rest of the Scenario for convenience, since the calculation takes some time.

In You can view the **Average Power Production** over a week or a month by clicking the blue buttons.